Model-Class Mismatch and Aggregation Hide Transcriptomic Signal in LINCS Drug Repurposing

Rishik Kondadadi
Eastview High School, *University of Minnesota*
Apple Valley, MN, USA
rishik.kondadadi@gmail.com, konda052@umn.edu

Dan McCreary
Kelly-McCreary Associates
Minneapolis, MN, USA
dan.mccreary@gmail.com

***Abstract*—A near-chance baseline is usually read as evidence that a modality carries no signal. For LINCS L1000 perturbational transcriptomics it is more often evidence about the pipeline. Measured the way most pipelines measure it, by averaging Level-5 signatures across all doses, cell lines and timepoints and fitting a random forest, drug-level expression recovers pharmacological class at AUROC 0.548 on Phase 2 (95% CI $[0.520, 0.577]$), close enough to chance to read as absence. That averaging cancels 64% of signature magnitude, and a tree ensemble is poorly matched to dense high-dimensional features. With both corrected it reaches 0.752 $[0.728, 0.776]$, and the corrected features recover known pharmacology unsupervised: HMGCR, the direct target of every statin, sits in the top ten of 978 landmark genes at $q \leq 0.018$. An independent release (LINCS Phase 1, 46,726 signatures) separates the general half of this diagnosis from the dataset-specific half. Model-class mismatch costs roughly 0.14 AUROC in seven of eight release-by-representation combinations and is a standing hazard. The aggregation effect reverses sign, with naive averaging the *best* representation on Phase 1 and the worst on Phase 2, so aggregation has to be settled per dataset. All of this came out of a negative control that failed decisively. A CLIP-style structure-expression aligner lost to a plain random forest on ECFP4 fingerprints in every class ($p \leq 0.027$), as did four stronger structural baselines and, at 0.751 against 0.917, the published alignment objective this recipe comes from ($p \leq 0.0039$). Correctly specified, the expression arm is still redundant with structure for these tasks ($+0.001$, p=0.15). We close with a self-audit of five defects found in our own pipeline, each of which produced a publishable-looking but incorrect number.**

***Index Terms*—drug repurposing, multi-modal learning, contrastive learning, LINCS L1000, transcriptomics, molecular fingerprints, negative results, reproducibility**

I. Introduction

Drug repurposing is an attractive target for machine learning: a repurposed compound skips much of the cost and time of a new molecular entity. A natural recipe, now widely instantiated, is to embed a drug's chemical structure and its measured transcriptomic response into a shared space with a contrastive objective, so that a disease signature can be matched against structurally plausible candidates.

Our starting point was a negative control for that recipe, which failed decisively. What made the failure worth following was that the diagnosis we first reached for it was wrong, and that correcting it recovered a modality we had been ready to write off. Our contributions:

1) **A modality that looks uninformative is recoverable.** Measured the way most pipelines measure it, drug-level L1000 expression sits at 0.548 on Phase 2 (95% CI $[0.520, 0.577]$), close enough to chance to read as absence of signal. Two measurement defaults produce that number. With both corrected it reaches 0.752 $[0.728, 0.776]$ (Section V), and the corrected features recover the canonical statin target without supervision (Section V-A).
2) **One half of the diagnosis is a general hazard, the other is dataset-specific.** On LINCS Phase 1 the model-class effect holds at $+0.09$ to $+0.16$ across representations and costs roughly 0.14 AUROC, while the aggregation effect reverses sign. Practitioners should treat model class as a standing hazard and determine aggregation on their own data (Section VI).
3) **The negative control is decisive and holds against the published objective.** Across six real pharmacological classes under a leak-free, drug-disjoint protocol, a random forest on raw ECFP4 fingerprints outperforms every contrastive variant we trained, in every class, and so do four stronger structural baselines. The published alignment objective of Finlayson et al. [1], reimplemented and tuned under the same protocol, loses in all six classes as well (Sections IV to IV-B).
4) **Correctly specified, the second modality is still redundant here.** With the expression arm properly constructed, blending does not improve on structure alone ($+0.001$, p=0.15), and the same holds on a second task family (Section VII).

We additionally report a self-audit (Section VIII) of five defects we found in our own pipeline over the course of this work. We include it because each defect produced a result that looked publishable, and because the sequence illustrates a failure mode that is difficult to catch from the outside: a chain of analyses that appear to be independent confirmations while all sharing a single upstream defect.

II. Related work

Contrastive alignment of molecules and transcriptomes. Pairing chemical structure with perturbational expression under a joint contrastive objective was introduced by Finlayson et al. [1], who also flagged generalisation across cell lines and compounds as unresolved. The CLIP formulation [2] that most such work adapts, and the alignment and uniformity diagnostics of Wang and Isola [3], are the two pieces of contrastive machinery we use directly. Subsequent work has applied contrastive objectives to drug-disease association tables and biomedical knowledge graphs, generally reporting favourable within-benchmark results.

Known problems with L1000 signal. Our diagnosis rests on findings that are individually established. Sirci et al. [4] report that a majority of CMap compounds produce weak transcriptional responses and that class recovery for those compounds is close to random, recommending that high-variability compounds be discarded. Lim and Pavlidis [5] document limited cross-version reproducibility of L1000 differential expression, with reproducibility depending strongly on differential-expression strength. What we add is the observation that these known properties interact with an ordinary preprocessing default to produce a chance-level result that looks like a modality limit.

Our own prior work in this line. One of us previously applied deep learning to epilepsy drug repurposing on LINCS data [6], using a temporal-diffusion architecture and reporting favourable results against published baselines. That work used the naive drug-level signature aggregation we identify here as problematic, and we have not re-evaluated it under the corrected pipeline. We single it out because the failure mode described here applies to our own earlier results as much as to the method we set out to test.

III. Setup

Data. LINCS L1000 Phase 2 (GSE70138) Level-5 signatures over 978 landmark genes, restricted to compound perturbations: 107,404 signatures over 1,768 compounds and 30 cell lines. Canonical SMILES were resolved via PubChem for 1,725/1,768 compounds (97.6%); all yielded valid 2048-bit ECFP4 fingerprints (RDKit, radius 2), retaining 103,989 signatures. Official LINCS signature QC metrics (`tas`, `distil_cc_q75`, `distil_ss`, `pct_self_rank_q25`) were obtained from the GEO supplementary `sig_metrics` file.

Labels. Six pharmacologically standard, mutually exclusive drug classes matched by compound name: antiepileptics (16 drugs), statins (5), SSRIs (6), NSAIDs (8), beta-blockers (6), antipsychotics (7). Positive counts are small, which is why we evaluate by repeated subsampling rather than a single split. A second task family, SIDER side-effect prediction, is used for replication (Section VII); 534 of our compounds map to SIDER, yielding 586 side-effect endpoints with at least 25 positive and 25 negative drugs each. The analyses in Section VII use subsets of 40 to 60 of these.

Protocol. Repeated Monte-Carlo *drug-disjoint* subsampling: for each of 10 seeds we hold out $\max(2, 35\%)$ of a class's positive drugs plus 150 random negative drugs; no drug's signatures ever appear in both the fitting and evaluation pools. Contrastive models use a CLIP-style symmetric InfoNCE objective between a structure encoder (ECFP4 $\rightarrow$ 256-d) and an expression encoder (978-d $\rightarrow$ 256-d), both 3-layer MLPs, learnable temperature; all in-batch pairs sharing a drug identity are treated as positives. Each pretraining regime is run with 3 independent seeds. Significance is Wilcoxon signed-rank, paired by subsampling seed after averaging over pretraining seeds.

IV. The negative control: structure alone beats cross-modal alignment

Table I reports downstream class recovery. A random forest on raw ECFP4 fingerprints is the strongest method in all six classes, beating the contrastive model by margins of 0.03 to 0.42 AUROC, significant in every case ($p \leq 0.027$). The raw-concatenation baseline (structure + expression features fed jointly to one model) is *worse* than structure alone in all six classes, an early hint of the interference we diagnose in Section V.

A. The result holds against stronger structural baselines

A random forest on ECFP4 is a simple comparator, and the natural objection is that we chose an easy target. We therefore added four stronger structural methods: a multi-representation random forest combining ECFP4 with MACCS keys and ten physicochemical descriptors; gradient boosting on the same representation; a Tanimoto-similarity k-nearest-neighbour scorer, the classic chemoinformatics approach; and a logistic model on a Tanimoto kernel, which is the structural analogue of the Gaussian interaction profile kernels that dominate the drug-disease association literature.

Combining fingerprints with MACCS keys and descriptors reaches 0.931, above the 0.917 of the ECFP4 model we report throughout. The kernel and nearest-neighbour methods land at 0.910 and 0.903. Gradient boosting at default settings is the one poor performer (0.581), which we attribute to overfitting a 2,225-dimensional representation with as few as five positive compounds; we report it rather than tuning it away.

The conclusion is unchanged and slightly strengthened. Our contrastive model does not merely lose to a convenient baseline. It loses to every reasonable structural method we tried, including the kernel formulation that this literature treats as standard.

B. The published alignment objective loses as well

The sharper objection is that our contrastive model is our own, so its failure might be a property of our implementation. We therefore reimplemented the alignment objective of Finlayson et al. [1], the work that introduced this recipe, from their Section 3.3: the adaptive-margin quadruplet loss over all four cross-modal pairs of two matched structure-expression pairs, a self-normalising expression encoder, and distance-weighted negative sampling over compounds. No code accompanies that paper. We hold the structural input at ECFP4 rather than their Chemprop D-MPNN so that what differs between the two aligners is the objective and not the featurisation, and we give it the same 25-epoch budget, the same splits and the same downstream probe. The margin

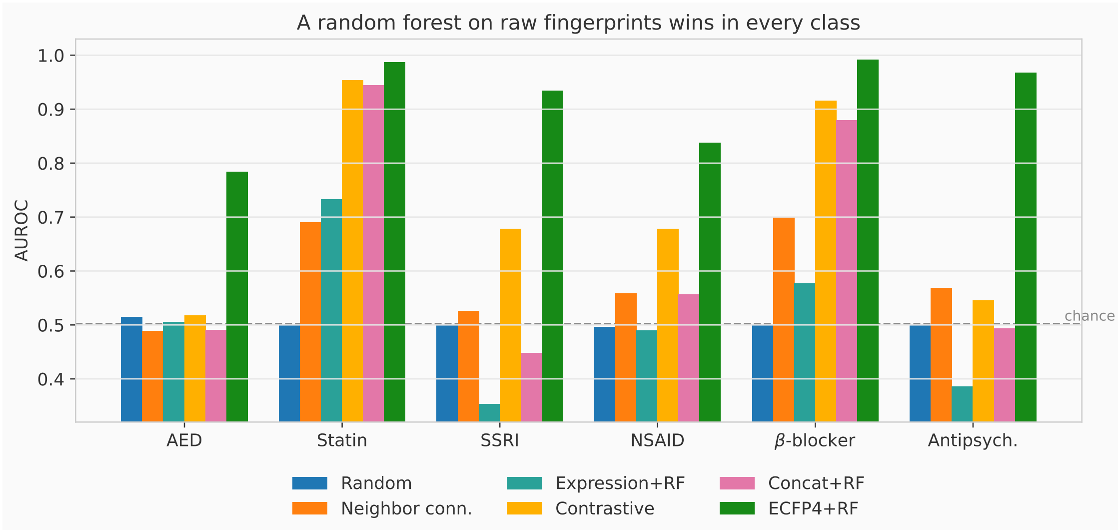

Fig. 1. Downstream class recovery across six pharmacological classes. The contrastive model is beaten by a random forest on raw ECFP4 fingerprints in every class. Note also that naively concatenating expression features to structure (*Concat+RF*) is *worse* than structure alone in all six classes.

TABLE I
DOWNSTREAM CLASS RECOVERY, MEAN AUROC OVER 10 DRUG-DISJOINT SUBSAMPLING SEEDS. CONTRASTIVE USES FULL-DATA PRETRAINING, CONCATENATED PROBE. A RANDOM FOREST ON RAW FINGERPRINTS WINS IN EVERY CLASS.

Method	AED	Statin	SSRI	NSAID	β-blocker	Antipsych.
Random	0.515	0.500	0.500	0.496	0.500	0.500
Neighbor connectivity (param.-free)	0.489	0.690	0.526	0.559	0.700	0.569
Raw expression + RF	0.506	0.733	0.353	0.490	0.577	0.386
Raw concat + RF	0.490	0.944	0.448	0.556	0.879	0.494
Contrastive (full, concat)	0.518	0.954	0.678	0.678	0.916	0.546
Raw ECFP4 + RF	**0.784**	**0.987**	**0.935**	**0.838**	**0.992**	**0.968**

TABLE II
STRONGER STRUCTURAL BASELINES, MEAN AUROC OVER SIX CLASSES AND 10 DRUG-DISJOINT SUBSAMPLING SEEDS. THE RICHER REPRESENTATION BEATS OUR ORIGINAL COMPARATOR, SO THE CONTRASTIVE MODEL LOSES TO A STRONGER TARGET THAN THE ONE REPORTED IN TABLE I.

Method	Mean AUROC
MACCS + descriptors + ECFP4, random forest	**0.931**
ECFP4, random forest	0.917
Tanimoto kernel, logistic regression	0.910
Tanimoto-similarity kNN	0.903
Multi-representation gradient boosting	0.581
Finlayson et al. quadruplet alignment [1]	0.751
Contrastive (full, concat)	0.715

parameters α and β were selected over a 3×6 grid by cross-modal retrieval MRR on a held-out fifth of the training compounds, never on the evaluation drugs.

It does better than our own aligner and still loses. Mean AUROC over the six classes is 0.751 against 0.715 for our CLIP-style model, so the published objective is the stronger of the two contrastive methods and our comparison is not resting on a weak stand-in. A random forest on ECFP4 beats it in all six classes, every one significant at $p \leq 0.0039$: AED 0.576 against 0.784, statins 0.930 against 0.987, SSRIs 0.650 against 0.935, NSAIDs 0.752 against 0.838, beta-blockers 0.901 against 0.992, antipsychotics 0.697 against 0.968. All four stronger structural baselines beat it too.

Two deviations from the published description are worth stating. The substitution of ECFP4 for the D-MPNN means we test their objective rather than their full system, so a graph encoder could recover some of the gap. And their negative sampler uses Wu et al.'s closed-form distance density, which assumes points spread uniformly on a sphere; drug-level mean expression is not, and the analytic form collapses the sampler onto each anchor's two nearest compounds, so we substituted the empirical density of the observed distances.

V. TWO MEASUREMENT DEFAULTS SUPPRESS THE EXPRESSION MODALITY

Our first reading of Table I was that L1000 expression simply carries no drug-level signal: expression-only averages 0.507 over the six classes, indistinguishable from the random control. We now believe that reading was wrong, and that the path to it is the most transferable finding in this paper.

Aggregation. To obtain one feature vector per drug we had averaged all of a compound's Level-5 signatures across doses,

TABLE III
THE FOUR CORNER CELLS OF THE AGGREGATION × MODEL GRID, RE-ESTIMATED AT 30 SUBSAMPLING REPEATS (n=180) WITH BOOTSTRAP CONFIDENCE INTERVALS. THESE ARE THE VALUES QUOTED IN THE ABSTRACT AND THROUGHOUT THE TEXT.

Aggregation	Model	AUROC	95% CI
Naive mean, all conditions	RF	0.548	$[0.520, 0.577]$
Naive mean, all conditions	LogReg	0.663	$[0.637, 0.689]$
High-dose 10 µM / 24 h	RF	0.635	$[0.606, 0.665]$
High-dose 10 µM / 24 h	LogReg	**0.752**	$[0.728, 0.776]$

cell lines, and timepoints. This is an unremarkable default. It is also destructive: the median norm of a drug's averaged signature is 11.3 versus 31.3 for its individual signatures, i.e. averaging cancels 64% of signature magnitude. The data are not the problem: 99.9% of compounds have at least one strongly active signature ($\geq$ 50 landmark genes at $|z| > 2$), and only 7.8% of individual signatures are weak in the sense of carrying fewer than 25 such genes. At 30 subsampling repeats, aggregation alone moves expression-only AUROC from 0.548 to 0.635 with the model held fixed at RF (Table III); the 6-repeat grid in Figure 2 puts the same shift at 0.525 to 0.619. Figure 3 shows the cancellation directly.

Model class. Independently, we had used a random forest for the expression arm, mirroring the structure arm. Random forests are poorly matched to dense, high-dimensional, low-sample expression features. Logistic regression on the *same* naively aggregated features reaches 0.663 $[0.637, 0.689]$ where the random forest reaches 0.548 $[0.520, 0.577]$. Structure, by contrast, is model-robust (RF 0.9171 vs. LogReg 0.9169), so the choice penalises only one arm.

The conjunction. Either choice alone leaves substantial signal; both together leave the modality barely distinguishable from chance. The best cell (high-dose aggregation + logistic regression) reaches 0.752 $[0.728, 0.776]$. Table III collects the four corner estimates, each at 30 subsampling repeats (n=180) with bootstrap intervals. Every corner value quoted in this paper comes from that table. The full grid in Figure 2 and the Phase 2 columns of Table IV use 6 repeats per cell and are intended to show the *pattern* across the design space rather than a precise level for any single cell. The three values this paper reports for the naive-mean random-forest cell are one estimator run for different numbers of repeats on the same seed sequence: 0.525 at 6 repeats, 0.507 at the 10 repeats behind Table I, and 0.548 at 30. That cell has a per-seed standard deviation of 0.060, so a 10-repeat mean carries a standard error near 0.019 and the three agree to within sampling noise. We quote the 30-repeat estimate throughout and use the shorter runs only where the comparison across cells, rather than the level of one cell, is the point.

This is probably not unique to us. Our naive-conjunction estimate (0.548) and published raw-signature baselines (median AUROC 0.507 for TS-978 and 0.485 for TS-12328 in the MoAble MoA benchmark [7]) both sit close to chance. We stress that this is agreement in range only. Our interval does not

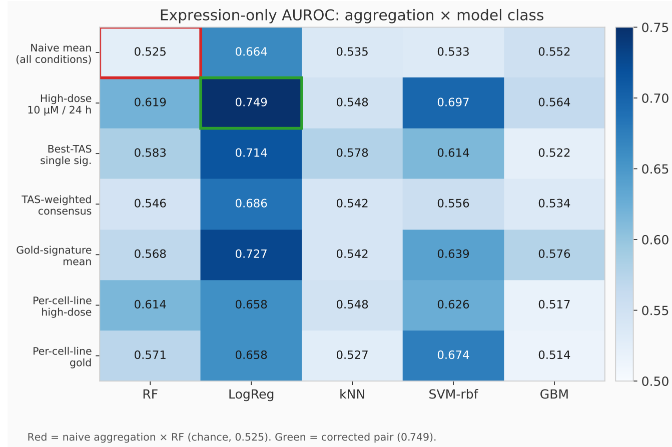

Fig. 2. Expression-only AUROC across aggregation strategies × downstream model classes (mean over six drug classes, 6 subsampling repeats per cell). This grid shows the *pattern*; the four corner cells are re-estimated at 30 repeats with bootstrap intervals in Table III. Neither axis alone explains the result: the naive-aggregation/random-forest cell (red) is the weakest in the grid, while correcting both axes (green) is among the strongest. Every other cell recovers partial signal, so the suppression requires the *conjunction*.

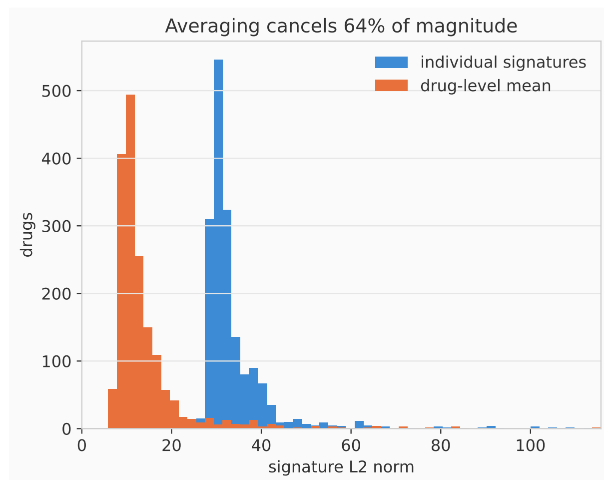

Fig. 3. Why naive aggregation destroys signal. Distribution over drugs of the L2 norm of individual Level-5 signatures (blue) versus the drug-level mean across all doses, cell lines, and timepoints (orange). Median magnitude falls from 31.3 to 11.3: opposing responses across conditions cancel.

contain 0.507, so we claim just that a near-chance published baseline is consistent with this class of artifact. Whether any particular published baseline arises from it is beyond what we can show. Prior work has separately established that a majority of CMap compounds show weak transcriptional response and that class recovery for those compounds is close to random [4], and that L1000 reproducibility depends strongly on differential-expression strength [5]. Our result is best read as a worked example of how preprocessing and model-selection defaults can manufacture a chance-level baseline where signal exists.

A hypothesis we tested and rejected. Given the above, we hypothesised that preprocessing choice would dominate model choice in explaining outcome variance. A variance decomposition over the 7×5 grid rejects this: between-model variance accounts for 73.3% of total variance versus 11.5% for between-aggregation, a ratio of $0.16\times$. We report this because we had intended it as the paper's headline and it did not survive contact with the data.

A. The corrected features recover known pharmacology

A reasonable objection to Section V is that our "corrected" pipeline might simply be overfitting a preprocessing choice to a metric. If the corrected expression features carry real biology, they should recover known drug mechanisms without being told what those mechanisms are. They do.

For each class we tested all 978 landmark genes for a difference in corrected response between class members and every other compound, using an exact two-sided Mann-Whitney U test, and kept the ten smallest p-values. Classes hold 5 to 16 compounds against roughly 1,700 others, so the exact null is the right one; the normal approximation is anti-conservative at these sizes. For statins the three largest effects among those ten are HMGCS1 ($\Delta = +3.66$, $p = 8.1 \times 10^{-5}$), RHOA ($+3.21$, 9.6×10^{-5}) and HMGCR ($+2.62$, 3.8×10^{-5}); a fourth sterol gene, ELOVL6, also appears ($+1.52$), and the remaining six carry effects below $|1.8|$. All ten survive Benjamini-Hochberg correction across the 978 genes, at $q \leq 0.018$. HMGCR encodes HMG-CoA reductase, the direct molecular target of every statin, and HMGCS1 sits immediately upstream in the same sterol-synthesis pathway. RHOA is the third expected hit rather than a stray one: statins deplete the geranylgeranyl pyrophosphate that RhoA needs for membrane anchoring. The positive sign is the expected direction: inhibiting HMGCR relieves sterol-mediated feedback and drives compensatory SREBP-2 transcription of both genes.

For SSRIs the five smallest p-values are EBP ($+1.72$), LIPA ($+1.42$), TMEM97 ($+1.69$), FDFT1 ($+2.17$) and HMGCS1 ($+3.08$). Four are sterol or lipid-metabolism genes. The fifth, TMEM97, is the sigma-2 receptor, and EBP is the sigma-1-associated sterol isomerase; both are established off-target binding partners of cationic amphiphilic drugs, a class to which most SSRIs belong. These do not survive correction (q from 0.109 to 0.257), so we read them as a consistency check on the features and not as discoveries.

Neither result was sought. We ran the analysis to characterise the features and found the canonical target of one class and a documented off-target pathway of another. This is a positive control on the corrected pipeline, and it is also the sharpest available evidence that the naive pipeline's chance-level result was destroying real biological signal rather than reporting its absence.

VI. Independent Replication: Half the Explanation Fails

A chance-level result invites a strong conclusion, so before drawing one we tested the explanation on an independent LINCS release. Phase 1 (GSE92742) is a separate experimental campaign with different cell lines, doses and timepoints.

TABLE IV
THE SAME AGGREGATION AND MODEL-CLASS GRID ON TWO INDEPENDENT LINCS RELEASES. EXPRESSION-ONLY AUROC, AVERAGED OVER SIX DRUG CLASSES, 6 SUBSAMPLING REPEATS PER CELL. PHASE 1 ROWS USE THE NEAREST AVAILABLE ANALOGUE OF EACH PHASE 2 REPRESENTATION. THE FOUR PHASE 2 CELLS IN THE TOP TWO ROWS ARE RE-ESTIMATED AT 30 REPEATS IN TABLE III.

	Phase 2		Phase 1	
Representation	RF	LogReg	RF	LogReg
Naive mean, all conditions	0.525	0.664	0.580	**0.730**
High-dose 10 µM / 24 h	0.619	**0.749**	0.543	0.686
Strongest single signature	0.583	0.714	0.548	0.513
Gold / exemplar signature mean	0.568	0.727	0.577	0.671

We extracted the 978 landmark genes for the 46,726 Phase 1 signatures covering 1,001 of our compounds, and confirmed the extraction with a positive control: on the corrected Phase 1 features HMGCR sits at $+1.53$ under statins against $+0.13$ elsewhere.

The model-class effect replicates. Logistic regression beats random forest by $+0.09$ to $+0.16$ in seven of the eight release-by-representation combinations of Table IV. The single exception is the strongest-signature representation on Phase 1 (-0.04), which is also the weakest representation overall. Whatever else is true, using a tree ensemble on dense high-dimensional expression features costs roughly 0.14 AUROC, and it costs it consistently.

The aggregation effect does not replicate, and reverses. On Phase 2, restricting to high-dose 24-hour signatures improved matters by $+0.09$ under both model classes. On Phase 1 the same restriction *hurts*, by -0.04 under both, and naive averaging over all conditions is the best representation tested. Our Phase 2 explanation, that averaging across heterogeneous conditions destroys signal through cancellation, cannot be the general account: the same operation on a different release of the same assay is harmless or mildly beneficial.

We do not have a confident explanation for the reversal. Phase 1 has broader dose and timepoint coverage per compound, so its naive average may pool more genuinely replicate measurements and fewer sub-threshold ones. That is a hypothesis and we have not tested it.

What we conclude, narrowly. Model-class mismatch is a robust hazard for this modality. Aggregation choice is consequential but release-dependent, and our chance-level Phase 2 number required both. A practitioner should not read this paper as saying "always restrict to high dose". The transferable instruction is to treat a near-chance expression baseline as a hypothesis about the pipeline, then test which part of the pipeline is responsible on their own data, because the answer differs between releases of the same assay.

This section exists because the replication contradicted us. We had written the conjunction claim before running it.

VII. A CORRECTLY SPECIFIED EXPRESSION ARM STAYS REDUNDANT WITH STRUCTURE

The obvious question is whether a properly specified expression arm becomes complementary to structure. It does not. With the best model chosen per modality (structure: RF on ECFP4; expression: logistic regression on high-dose signatures), a fixed 50:50 probability blend yields:

	Structure	Expression	Blend
Mean AUROC (pooled, n=120)	0.9160	0.7514	0.9173

a gain of $+0.0013$ ($p = 0.15$, 27/120 wins). Expression carries real signal ($0.751 \gg$ chance) but that signal is largely subsumed by structure for this task. Per-class gains range from -0.004 to $+0.011$ and are individually non-significant.

Replication on a second task family. Pharmacological class is largely structure-determined by construction, so we repeated the analysis on SIDER side-effect prediction, where prior work reports expression outperforming structure [8]. With nested (leak-free) blend-weight selection over 60 side effects, structure reaches 0.589, expression 0.530, and the blend 0.573; blending *hurts* (-0.017, $p = 1.8 \times 10^{-4}$). Gold-signature per-cell-line features, the closest replication of [8] we can construct from Phase 2 data, do not change the ordering: expression reaches 0.543 on those same endpoints against 0.589 for structure. Nor does the independent release. On Phase 1, across 546 endpoints and 428 compounds, structure reaches 0.600 against 0.512 for strongest-signature expression features and 0.526 for exemplar signatures, a deficit of -0.09 ($p = 1.4 \times 10^{-63}$, with expression ahead on only 87 of the 546 endpoints). We did not replicate the published result on either release. We flag it as an open discrepancy rather than a refutation, since we lack the Phase 1 QC filtering and the GO-enrichment transform used in that work.

One cell where blending does help. Restricting to transcriptionally responsive compounds (strongest signature at TAS $\geq$ 0.3, following Sirci et al. [4]) and using gold-signature per-cell-line features, the fixed blend gains $+0.021$ over structure on drug-class recovery ($p = 0.017$, 40 repeats). We report it because it cuts against our conclusion, and we do not read much into it. The same cell with a random-forest expression arm gains $+0.026$ at $p = 0.60$. It is one of a dozen blend comparisons we ran across representations and compound subsets, so it does not survive correction for that count. And on SIDER the same restriction with the same features makes blending worse (-0.012, $p = 0.027$). We take the blend gain to be representation- and task-dependent scatter around zero rather than a recoverable complementarity.

We also caution that an earlier version of this analysis appeared to show a $+0.008$ gain; that gain was entirely an artifact of selecting the blend weight on test predictions, and vanished under nested selection.

VIII. SELF-AUDIT: FIVE DEFECTS, EACH OF WHICH PRODUCED A PUBLISHABLE RESULT

Over the course of this study we found five defects in our own pipeline. We list them because each yielded a plausible, presentable result, and because their interaction is the substantive methodological warning of this paper.

1) **Zero-vector embeddings.** In the class-holdout regime, drug-level expression embeddings were aggregated only over signatures visible to that regime, so held-out drugs received all-zero embeddings, which are trivially separable. This produced apparent AUROC of 1.000 ± 0.000. Caught only because the value was implausibly perfect.
2) **Single-seed regime comparison.** The full and class-holdout regimes were initially compared at one pretraining seed each, confounding the regime effect with seed variance. Fixed by running three seeds per regime.
3) **Naive signature aggregation** (Section V).
4) **Mismatched model class for the expression arm** (Section V).
5) **Zero-filled representations**, found while preparing the code release for this paper. Every aggregation that selects a subset of a compound's signatures left an all-zero vector behind for compounds with no signature meeting its criterion: 49 of 1,725 for the high-dose subset, 50 for the gold subset. This is defect 1 again, in a different part of the pipeline, and it favoured the random forest, which can split on the all-zero pattern: the high-dose random-forest cell fell from 0.641 to 0.619 once each compound fell back to its own naive mean, while the logistic cell rose from 0.733 to 0.749. Every number in this paper uses the corrected fallback. The conclusions did not change, which is the only reason this defect is a footnote rather than a retraction.

Defects 3 and 4 are the instructive pair. Once defect 3 had convinced us that the modality was uninformative, we then ran five follow-up analyses (a nonlinear probe, combined raw+learned features, an untrained-encoder control, a few-shot sweep, and a second task family), all of which agreed. They agreed because they all consumed the same defective features. *Mutual confirmation among analyses sharing an upstream input is not independent evidence*, and from the outside this pattern is nearly indistinguishable from a well-controlled negative result. In our case the error was surfaced by an external prompt to check the domain literature, which produced an exact numerical match to a known artifact.

IX. RECOMMENDATIONS

For practitioners combining chemical structure with L1000 transcriptomics: (i) do not aggregate Level-5 signatures by unweighted mean across dose, cell line, and timepoint; prefer high-dose/fixed-timepoint subsets, per-cell-line feature blocks, or QC-filtered consensus; (ii) select the downstream model class per modality, as structure and expression have very different geometry; (iii) treat any expression baseline landing near 0.50 as a preprocessing hypothesis to be excluded before it is reported as a modality limit; and (iv) when several analyses agree, verify that they do not share an upstream transformation.

For this task family specifically, our results indicate expression is informative but largely redundant with structure; we do not claim this transfers to tasks where structure is less determinative, and the unreplicated Wang et al. [8] result is a live counterexample worth resolving.

X. Conclusion

A near-chance baseline is the point at which most people stop measuring a modality. On LINCS L1000 that is usually the wrong moment to stop. Drug-level expression sits at 0.548 under two ordinary measurement defaults, averaging Level-5 signatures over every experimental condition and reusing the structure arm's model class, and at 0.752 once both are corrected. The corrected features are not an artifact of tuning a preprocessing choice against a metric: they put HMGCR, the direct molecular target of every statin, in the top ten of 978 landmark genes at $q \leq 0.018$, alongside HMGCS1 and RHOA.

Half of that diagnosis generalises. On an independent LINCS release, model-class mismatch costs roughly 0.14 AU-ROC in seven of eight release-by-representation combinations, so a tree ensemble on dense expression features is a standing hazard. The aggregation effect reverses: the restriction that helped on Phase 2 hurts on Phase 1, where naive averaging is the best representation we tried. We report the half that failed because we had already written the conjunction claim when the replication contradicted it. The transferable instruction is to treat model class as a hazard everywhere and to settle aggregation on the data at hand.

The negative control that started this stands. A CLIP-style aligner lost to a random forest on fingerprints in all six disease areas, four stronger structural baselines beat it as well, and so does the published objective this recipe comes from once reimplemented and tuned under the same protocol. Correctly specified, the expression arm remains redundant with structure for these tasks. Recovery and complementarity are different problems, and this paper solves only the first.

Limitations

One published objective, not every published system. Section IV-B tests the alignment objective of Finlayson et al. [1] under our protocol, and it loses in all six classes. That closes the obvious version of this gap, but not all of it. We reimplemented their objective with an ECFP4 encoder rather than their Chemprop D-MPNN, so a graph structure encoder could recover part of the margin, and we have not run the drug-disease contrastive systems built on knowledge graphs, which consume different inputs and would need a different evaluation. Our claim is that two contrastive alignment objectives, one of them the published one, lose to a random forest on fingerprints under this protocol.

One assay. Results come from LINCS L1000 Phase 2 with replication on Phase 1 (Section VI), so they generalise across releases but not beyond the assay. The SIDER replication changes the labels while keeping the same compounds and measurements, varying the task rather than the data source. Whether the conjunction reproduces on CMap-independent expression data, or under a different structural representation, is untested. The Phase 1 QC fields used by some prior work were unavailable, so the replication uses unfiltered signatures.

Small positive classes. Class sizes of 5–16 drugs bound per-class power, mitigated by repeated subsampling and by effects consistent across classes. Membership comes from name matching, not measured efficacy.

Code and data availability

Every number in this paper regenerates from the public LINCS releases using the scripts at https://github.com/edanmn/maidrs-2026-modality, which also holds the result log or CSV behind each one and a README mapping every table and claim to its source. Each figure script prints the values the paper quotes from it, so those can be checked against a fresh run rather than read off the figure. The raw LINCS releases (GSE70138, GSE92742) and SIDER 4.1 are public but too large to redistribute; the scripts that build our dataset from them are included.

References

[1] S. G. Finlayson, M. B. A. McDermott, A. V. Pickering, S. L. Lipnick, and I. S. Kohane, "Cross-modal representation alignment of molecular structure and perturbation-induced transcriptional profiles," in *Pacific Symposium on Biocomputing*, vol. 26, 2021, pp. 273–284.

[2] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. 38th Int. Conf. Machine Learning*, PMLR 139, 2021, pp. 8748–8763.

[3] T. Wang and P. Isola, "Understanding contrastive representation learning through alignment and uniformity on the hypersphere," in *Proc. 37th Int. Conf. Machine Learning*, PMLR 119, 2020, pp. 9929–9939.

[4] F. Sirci, F. Napolitano, S. Pisonero-Vaquero, D. Carrella, D. L. Medina, and D. di Bernardo, "Comparing structural and transcriptional drug networks reveals signatures of drug activity and toxicity in transcriptional responses," *npj Systems Biology and Applications*, vol. 3, p. 23, 2017.

[5] N. Lim and P. Pavlidis, "Evaluation of connectivity map shows limited reproducibility in drug repositioning," *Scientific Reports*, vol. 11, no. 1, p. 17624, 2021.

[6] R. Kondadadi, "Causal temporal diffusion networks for drug repurposing in epilepsy," *Briefings in Bioinformatics*, vol. 26, suppl. 1, p. i42, Dec. 2025, conference abstract, doi: 10.1093/bib/bbaf631.074.

[7] G. Jang, S. Park, S. Lee, S. Kim, S. Park, and J. Kang, "Predicting mechanism of action of novel compounds using compound structure and transcriptomic signature coembedding," *Bioinformatics*, vol. 37, suppl. 1, pp. i376–i382, Jul. 2021, doi: 10.1093/bioinformatics/btab275.

[8] Z. Wang, N. R. Clark, and A. Ma'ayan, "Drug-induced adverse events prediction with the LINCS L1000 data," *Bioinformatics*, vol. 32, no. 15, pp. 2338–2345, 2016.